

Electron-Ion Collider, Brookhaven National Laboratory			
Doc No. EIC_PSG_SOW-003	Author: Dominic Prevosto	Effective Date: [Insert here]	Review Frequency: N/A
SOW: Statement of Work for the EIC Rack Enclosures			Revision: 00

STATEMENT OF WORK
for the
EIC Rack Enclosures

Quality Classification: A-3

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REVISION HISTORY

Revision #	Effective Date	Summary of Change
00		Initial Release

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LIST OF ACRONYMS

AHJ	Authority Having Jurisdiction
ASME	American Society of Mechanical Engineers
BNL	Brookhaven National Laboratory
BSA	Brookhaven Science Associates
CAD	Computer Aided Design
EIC	Electron-Ion Collider
ESH	Environment, Safety and Health
FDR	Final Design Review
ISO	International Organization for Standardization
NRTL	Nationally Recognized Test Laboratory
PDR	Preliminary Design Review
PO	Purchase Order
QA	Quality Assurance
SOW	Statement of Work

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EIC Rack Enclosure

1 SCOPE

This Statement of Work (SOW) will be the basis for the procurement of Rack Enclosures to be used at the Electron-Ion Collider (EIC) at Brookhaven National Laboratory (BNL) managed by Brookhaven Science Associates (BSA). This document outlines the Contractor's responsibilities and obligations necessary for completing the requirements as set forth herein.

1.1 Background

BNL was chosen as the building site for the EIC, a one-of-a-kind nuclear physics research facility. The EIC will be a discovery machine for unlocking the secrets of the "glue" that binds the building blocks of visible matter in the universe. The machine design is based on the existing and highly optimized RHIC Ion-Ion collider. It will consist of two intersecting accelerators, one producing an intense beam of electrons, the other a beam of protons or heavier atomic nuclei which are steered into collisions. Beyond sparking scientific discoveries in a new frontier of fundamental physics, the Electron-Ion Collider will trigger technological breakthroughs that have broad-ranging impact on human health and national challenges. The facility, components and systems procured for EIC, are designed to meet requirements to ensure high reliability and availability for the user program.

The EIC rack enclosures will be housing many different devices for a variety of groups at BNL. These enclosures will be holding power supply equipment, RF equipment, controls equipment, and so forth.

2 APPLICABLE DOCUMENTS

In the event of a conflict between the Technical Documentation and this SOW, the Contractor shall immediately notify the BSA Contractual Representative who will in each instance determine which document takes precedence and advise the Contractor accordingly. Failure to notify BSA of a document conflict shall not relieve the Contractor's responsibility to ensure full compliance to all requirements.

The following documents are an integral part of the SOW; the applicable revision level will be the latest that is in effect at the time of award:

Document Number	Document Title
EIC-PSG-SPC-005	Technical Specification for the EIC Rack Enclosures
ASME Y14.100	Engineering Drawing Practices

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BNL-QA-101	Brookhaven National Laboratory Supplier Quality Assurance Requirements
IEEE 1202	FT4 Flame Test
ASME B30.26	Rigging Hardware
AME BTH-1	Design of Below-the-hook Lifting Devices Category “A”
UL508	Safety for Industrial Control Panels

3 REQUIREMENTS

In accordance with the applicable specifications, referenced documents, and instructions as defined in this Statement of Work (SOW), the Contractor shall be responsible for designing, manufacturing, inspecting, testing, and delivering the rack enclosures.

3.1 Contractor Responsibilities

3.1.1 Technical Performance

In accordance with this SOW, BSA technical specification EIC-PSG-SPC-005, and all referenced documents, the Contractor shall furnish all equipment, materials, tools, facilities and labor to perform the work necessary to design and analyze, manufacture and test the EIC Rack Enclosures. The Contractor shall ensure that the resultant design & finished product meets the performance parameters and criteria in accordance with BSA specification EIC-PSG-SPC-005 and all other referenced documents.

3.1.2 Rack Package Deliverables

For each type of rack performance package, the Contractor shall:

- Complete the final design, all required analysis, and delivery of drawings and 3D computer aided design (CAD) models
- After design approval, deliver a first article rack for evaluation by the BSA technical team
- After approval of the first article rack performance, complete the fulfillment of the remainder of the order

3.1.3 Tooling/Fixtures/Test Equipment

The Contractor shall be responsible for supplying all materials, including (but not limited to) the construction of all tools, jigs, fixtures, and test equipment required to meet the requirements of this SOW and the referenced technical drawings and specification(s).

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3.1.4 Materials

The Contractor shall have an effective storage and age control system for items where acceptability is limited by the age or manner of storage of the item.

3.2 Engineering Design and Manufacturing Requirements

3.2.1 Detailed Analysis Requirements

The Contractor shall perform the following analysis to ensure the equipment meets the requirements of BSA specification EIC-PSG-SPC-005. Models and results shall be provided electronically. If ANSYS or other FEA codes are used, results shall be provided in their native or equivalent, format, as well as in hard copy form.

3.2.1.1 Stress and Deflection Calculations

The Contractor shall provide comprehensive mechanical calculations and supporting documentation demonstrating that the lifting lugs are safe and the lugs and lug attachment to the rack are designed with a factor of safety of at least three (3) when supporting a fully populated rack. Similarly, the Contractor shall document that the internal mount rails and the general overall rack structure can safely carry the fully populated load.

3.2.1.2 Thermal Calculations

The Contractor shall provide comprehensive thermal calculations and supporting documentation demonstrating that the proposed rack enclosure can meet the specified cooling requirements under the expected operating conditions. This analysis shall include heat load estimates, airflow modeling, component temperature predictions, and confirmation that the enclosure maintains all power supplies and internal equipment within their allowable temperature limits at maximum specified load. All assumptions, methodologies, and safety margins used in the thermal analysis shall be clearly stated.

3.2.1.3 Design Reviews and Design Documentation

The Contractor shall hold a preliminary design review (PDR) and a final design review (FDR). Depending on the complexity of the rack type, the Contractor may request forgoing a PDR and moving directly to a FDR, subject to BSA Technical Representative approval. A design report to be approved by the BSA Technical Representative, in writing, shall be provided at each design review milestone.

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3.3 Management

3.3.1 Progress Communications

Within 2 weeks after award of contract, biweekly program, technical, and progress telecommunications and/or meetings between the Contractor and BSA shall be held at a mutually scheduled time. The discussions shall include the Contractor's progress, technical and contractual questions, presentations of analysis or testing results, design reviews, value engineering, troubleshooting, material status, tooling status, resources, and manufacturing issues.

3.3.2 Performance Reporting

Throughout the life of the Contract, the Contractor shall supply a written report by the fifth of every month to the BSA Technical Representative and Contractual Representative clearly detailing progress through the prior month with respect to the program. This shall include milestone charts, detailed progress, open items, problems, recommended solutions, and risk mitigation.

3.3.3 Report/Documentation Format

All reports submitted shall include an introduction, body of the documents, and a conclusion. Large documents shall include a table of contents. For calculations, equations shall be shown with values defined for each variable, so the analysis can be reviewed. Diagrams shall be included for clarity and labels shall be consistent with design drawings for specific items, such as vessel penetrations. All documentation provided by the Contractor shall be in English.

The Contractor shall provide an electronic copy (BSA-identified FTP, DVD or USB data storage device) of all documentation required by this SOW. Contractor manufacturing drawing Computer-Aided Design files of all components and assemblies shall be in one of these formats: STP, SAT, IGES, IAM, or IPT.

3.3.4 Engineering Drawing and 3D Model Practices

All Contractor-generated engineering drawings shall be in accordance with ASME Y14.100, Engineering Drawing Practices. All dimensions and tolerances shall be U.S. Customary units with metric (SI) units as a secondary notation.

All Contractor-generated 3D CAD part and assembly models, regardless of the software package used, shall be made available as neutral STEP files for importing into other design software

3.3.5 Documentation Required for Fabrication/Production

Prior to the start of first-article fabrication the Contractor shall provide:

1. Detailed 3D CAD model of each type of production rack including the cooling system and any other integrated electrical components. The model provided shall be of sufficient detail for the BSA Technical Representative to evaluate the installation of full racks in various areas around EIC, and to evaluate the installation of components into the rack.

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2. Documentation covering the certification, safety, and operation of integrated cooling systems and other electrical equipment.

3.3.6 Records

The Contractor shall retain objective evidence, including records, of the inspections and tests performed during manufacturing, testing, inspecting, preserving, packaging, and preparation for shipment of procured items. These records shall be made available to BSA for review upon request. These records shall be maintained for a minimum of three (3) years, unless otherwise specified in the procurement documentation, after the completion of the Purchase Order (PO)/Contract.

3.4 Environment, Safety and Health (ESH)

The Contractor shall design and build all equipment in accordance with ESH requirements in the appropriate specifications, codes, standards, and other documents listed in section 2 of this SOW.

- Fire
 - The Contractor shall ensure that any electrical equipment or wiring minimizes combustibility by the use of flame-resistant materials.
 - All cables to be installed in a cable tray shall be rated for Cable Tray use. The cable shall have the TC (Tray Cable) and NRTL markings. If a Contractor elects to use a cable not marked for Tray Cable use, then the Contractor shall use a cable that will meet IEEE 1202 or equivalent approved by BSA in writing. BSA shall approve the use of all non-NRTL marked tray cables in advance of the Contractor ordering the cable.
 - All other cables shall be Low Smoke Zero Halogen (LSZH)
- Material Handling
 - Lifting hardware must meet BSA safety requirements, and as such shall be designed according to ASME B30.26, "Rigging Hardware", and ASME BTH-1 (Design of Below-the-Hook Lifting Devices, Design Category "A").
- Electrical Safety
 - All electrical equipment, material, and systems delivered to BNL shall be certified, listed, or labeled by a Nationally Recognized Testing Laboratory (NRTL). The CE mark is NOT a recognized NRTL certification mark. (For a listing of OSHA-recognized NRTLs, refer to <http://www.osha.gov/>)
 - For electrical equipment, material, and systems which are not NRTL listed, the Supplier shall determine the equipment to be safe for its intended use. The determination must be made on the basis of test data. The determination and test data documents shall be made available to BSA prior to or upon delivery for review and acceptance by the applicable BSA Authority Having Jurisdiction (AHJ).
 - All electrical connections ≥ 50 volts shall be covered to comply with UL 508 so that all uninsulated charged surfaces protect workers from the hazards of electrical

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shock. Plexiglas® is not allowed; Lexan® may be used. Electrical insulation must be Low Smoke, Zero Halogen and Fire Retardant.

3.5 Marking and Serialization

Where labeling is required, the position of serialization and/or labeling of the identification number/nameplate shall be agreed upon between the Contractor and the BSA Technical Representative prior to acceptance of final design.

3.6 Packaging

In preparation for shipping, the Rack Enclosures shall be covered or wrapped for protection against weather elements, dust, and damage to projections, shock loads, impacts, and rubbing. The Rack Enclosures shall be properly braced and cushioned within the packing so that they will not shift during handling and shipment. The method of shipping will be specified by BSA one month prior to delivery. The container shall be appropriately labeled with the listing of contents, destination, gross weight, location of center of gravity, Purchase Order/Contract number, and contact information. Each container shall have G-force shock indicators. Each container shall be palletized and include provisions to permit access for a forklift to lift the container.

4 QUALITY ASSURANCE

4.1 Quality System and Facility Access

The Contractor shall maintain and apply an effective and documented quality assurance system for the design, manufacture, and testing of all equipment provided. The system shall meet the requirements of the ISO-9001 Standard, "Quality Management Systems Requirements", and shall provide the processes to ensure that each component produced for acceptance conforms to the requirements of this document and the applicable technical documentation. The Contractor shall supply the necessary documentation to demonstrate this compliance. Post-award surveys may be required and shall be supported by the Contractor. In addition, BSA representatives shall have access to the Contractor's facility during agreed-upon fabrication, inspection and testing processes. The Contractor shall provide ten (10) working days' notice in advanced of the agreed-upon witness points.

As set forth in BSA Form BNL-QA-101, "Brookhaven National Laboratory Supplier Quality Assurance Requirements," as amended herein, forms a part of this SOW. Specifically, this includes the following sections: 1, 2, 3, 3.1, select between 3.1.1 to 3.1.4 , 3.2 to 3.8, 3.21, 3.22, 4.0, 4.1, 4.2, 4.3 , 4.4 , select one or more between 4.4.1 to 4.4.3, 4.5, 4.6, 4.7, select between 4.7.1 to 4.7.3 as applicable, 4.10, select between 4.10.1 to 4.10.4 as applicable, 4.11, 4.13, 4.14, 4.16, 4.18, select one or more between 4.18.1 to 4.18.4, 4.19, 4.21, 4.34, and 4.39.

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BNL-QA-101 Section 2.0 Definitions use the terms Buyer and Seller. In the context of this SOW Buyer is referred to as “BSA” and Seller is referred to as “Contractor”.

BNL-QA-101 Clause 4.5 “Witness Points” is amended to require the contractor to provide BSA with ten (10) working days’ notice in advance of reaching such witness points during the manufacturing and test cycle of each item and that the BSA Technical Representative shall review and provide written approval prior to shipment.

BNL-QA-101 Clause 4.39 “Records” is amended to require the Contractor to maintain records for a minimum of five (5) years.

4.2 Configuration Management

The Contractor shall establish and maintain a configuration control system to ensure that all end items are of the proper configuration, and that all approved configuration changes are incorporated at the specified effectivity points. Records verifying the configuration of each item shall be maintained for a minimum of five (5) years from the date of acceptance of the equipment by BSA. The Contractor shall immediately notify the BSA Contractual Representative of any significant changes (those that may affect form, fit, function, reliability, safety, or interchangeability) in design, fabrication methods, material or processing.

For changes in configuration, a request for change approval shall be submitted to BSA for consideration. Changes in configuration may be made by the Contractor only with the written approval of BSA.

4.3 Inspections and Test Requirements

Unless otherwise specified in the PO/Contract, the Contractor shall be responsible for the performance of all inspections and tests specified herein. Except as otherwise specified in the PO/Contract, the Contractor shall utilize facilities suitable for the performance of the inspection and test requirements, specified herein, which are acceptable to BSA. BSA reserves the right to re-perform any of the inspections and/or tests set forth in this document or referenced specifications where deemed necessary to ensure that Equipment conforms to prescribed requirements at no cost to the Contractor. BSA and/or their representative shall have the option to witness all of the tests defined in this SOW.

The systems shall be tested by the Contractor at the Contractor’s facility, the Contractor’s subcontractor’s facility, or a pre-approved BSA collaborator prior to shipment. BSA or its designee retains the right to witness all testing. Final acceptance of the units shall take place following the successful completion of final testing at BNL. The time period for the final testing at BNL shall be within 90 days after receipt of the units at BNL.

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4.4 Repairs

If failures occur or repairs are required to an assembly or any component during manufacturing or testing that affects form, fit, function, reliability, safety, or interchangeability, the process shall stop and BSA shall be immediately notified. BSA at their discretion may ask for retesting, additional review and evaluation to determine root causes, analyses of failure modes(s) or proposals for repair. Proposals for repair shall include adequate information for evaluating the repair, testing the effectiveness of the repair, and methods to prevent damage to the existing work. No proposed repair shall be initiated without approval of BSA. No repair shall be considered successfully accomplished without inspection and written approval by BSA. All repairs shall be at the expense of the Contractor. In addition, the Contractor shall not use nonconforming materials via use-as-is disposition without written approval of BSA.

5 DELIVERABLES

The Contractor shall supply the following:

Item	Deliverable	Due	BSA Approval
1	Program Plan	2 weeks after award of contract	Yes
2	Technical and Progress Reporting	2 weeks after award of contract/ bi-weekly thereafter	No
3	Draft Preliminary Design Review Report	10 working days prior to PDR	No
4	Final Preliminary Design Review Report	10 working days after PDR	Yes
5	Draft Final Design Review Report	10 working days prior to FDR	No
6	Final Design Review Report	10 working days after FDR	Yes
7	Delivery of First Article	After FDR and as set forth in schedule	Yes
8	Delivery of unit 3D CAD model, and safety and operation documentation	After FDR and as set forth in schedule	Yes
9	End Item Documentation	As set forth in schedule	Yes
10	Delivery of Balance of Equipment	As set forth in schedule	Yes

6 SCHEDULE

The following schedules are for planning purposes only. Schedule requirements set forth in the Contract will take precedence.

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Milestone	Calendar weeks after award
Preliminary Design Review	Start + 4 weeks
Final Design Review	PDR6 weeks
Delivery of CAD models and operating instructions	FDR+6 weeks
Delivery of First Article	FDR+14 weeks
Delivery of End Item Documentation	First Article + 2 weeks
Beginning of Delivery of Balance of Equipment to BNL	End Docs + 14 weeks

7 VALUE ENGINEERING

The Contractor is encouraged to make recommendations for changes that might lead to an improvement in performance, reliability, quality, safety, or reduction in cost. Simplicity in design and manufacture and an improvement in the performance and reliability beyond the requirements of this specification are objectives which shall be considered in the Rack Enclosures. Where it appears a substantial improvement in simplicity of design, performance, reliability will result from the use of materials, parts and processes other than those specified, it is desirable that their use be investigated. When investigations point to advantages that may be realized, the recommendation for change shall be presented to BSA for review.